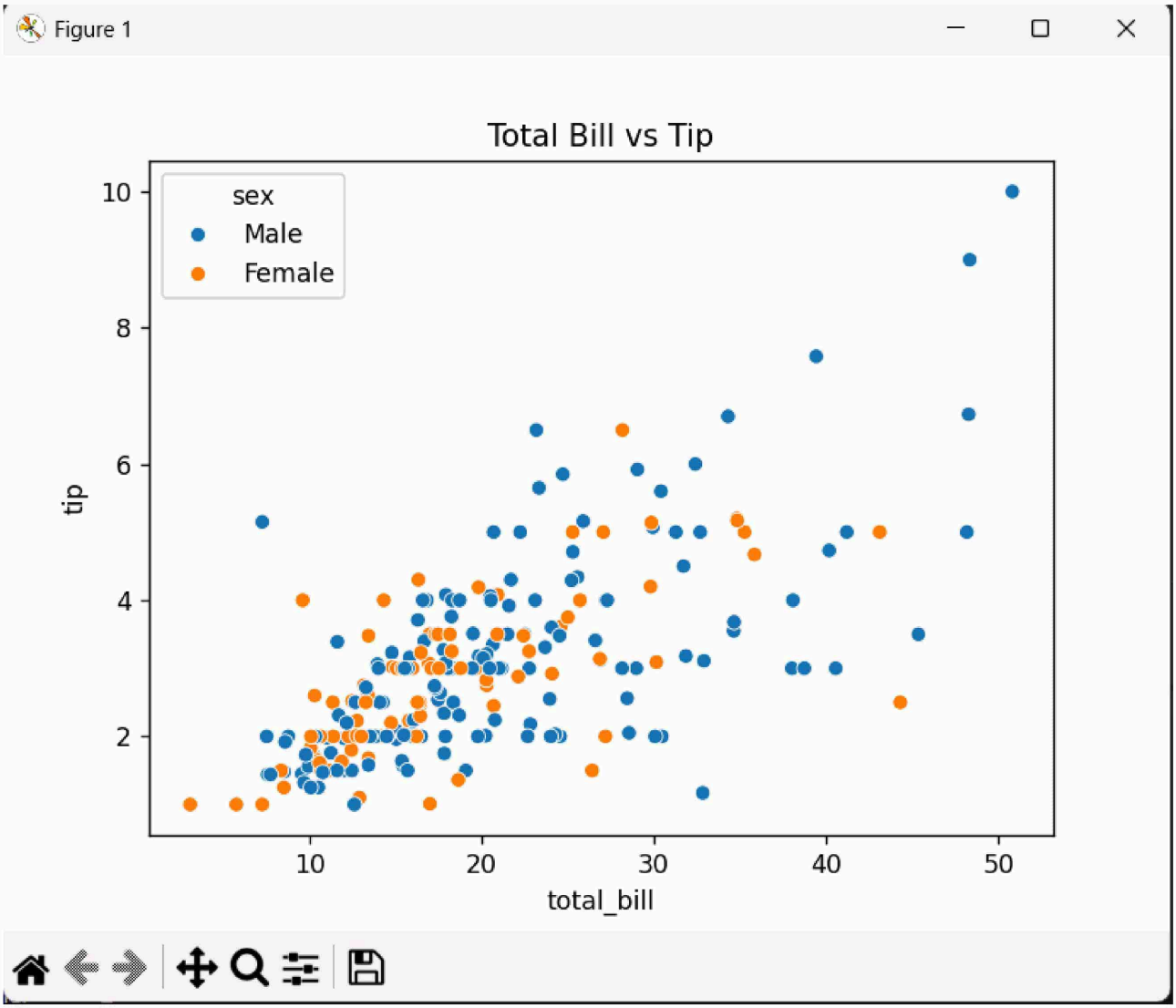
```
# Task1
import seaborn as sns
import matplotlib.pyplot as plt

tips = sns.load_dataset("tips")

print(tips.head())
print(tips.info())

# Task2
sns.scatterplot(data=tips, x="total_bill", y="tip", hue="sex")
plt.title("Total Bill vs Tip")
plt.show()

avg_tip = tips.groupby("day")["tip"].mean().reset_index()

sns.lineplot(data=avg_tip, x="day", y="tip", marker="o")
plt.title("Average Tip per Day")
plt.show()

sns.barplot(data=tips, x="day", y="total_bill")
plt.title("Average Total Bill by Day")
plt.show()

sns.histplot(tips["total_bill"], kde=True)
plt.title("Distribution of Total Bill")
plt.show()

# Task3
fig, axes = plt.subplots(2, 1, figsize=(6,8))

sns.scatterplot(data=tips, x="total_bill", y="tip", ax=axes[0])
axes[0].set_title("Scatter Plot")

sns.histplot(tips["total_bill"], ax=axes[1])
```

```
axes[1].set_title("Histogram")

plt.tight_layout()
plt.show()

# Task4
corr = tips.corr(numeric_only=True)

sns.heatmap(corr, annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()

# Task5
titanic = sns.load_dataset("titanic")

sns.countplot(data=titanic, x="survived")
plt.title("Survival Count")
plt.show()

sns.histplot(titanic["age"], kde=True)
plt.title("Age Distribution")
plt.show()

sns.countplot(data=titanic, x="sex", hue="survived")
plt.title("Survival by Gender")
plt.show()

corr = titanic.corr(numeric_only=True)

sns.heatmap(corr, annot=True, cmap="viridis")
plt.title("Titanic Correlation Heatmap")
plt.show()
```

Outputs:

```
PS D:\Internship\Day22> python task1-5.py
   total_bill   tip     sex smoker  day    time  size
0       16.99  1.01  Female     No  Sun  Dinner     2
1       10.34  1.66    Male     No  Sun  Dinner     3
2       21.01  3.50    Male     No  Sun  Dinner     3
3       23.68  3.31    Male     No  Sun  Dinner     2
4       24.59  3.61  Female     No  Sun  Dinner     4
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 244 entries, 0 to 243
Data columns (total 7 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   total_bill  244 non-null    float64
 1   tip         244 non-null    float64
 2   sex         244 non-null    category
 3   smoker      244 non-null    category
 4   day         244 non-null    category
 5   time        244 non-null    category
 6   size        244 non-null    int64
dtypes: category(4), float64(2), int64(1)
memory usage: 7.4 KB
None
```



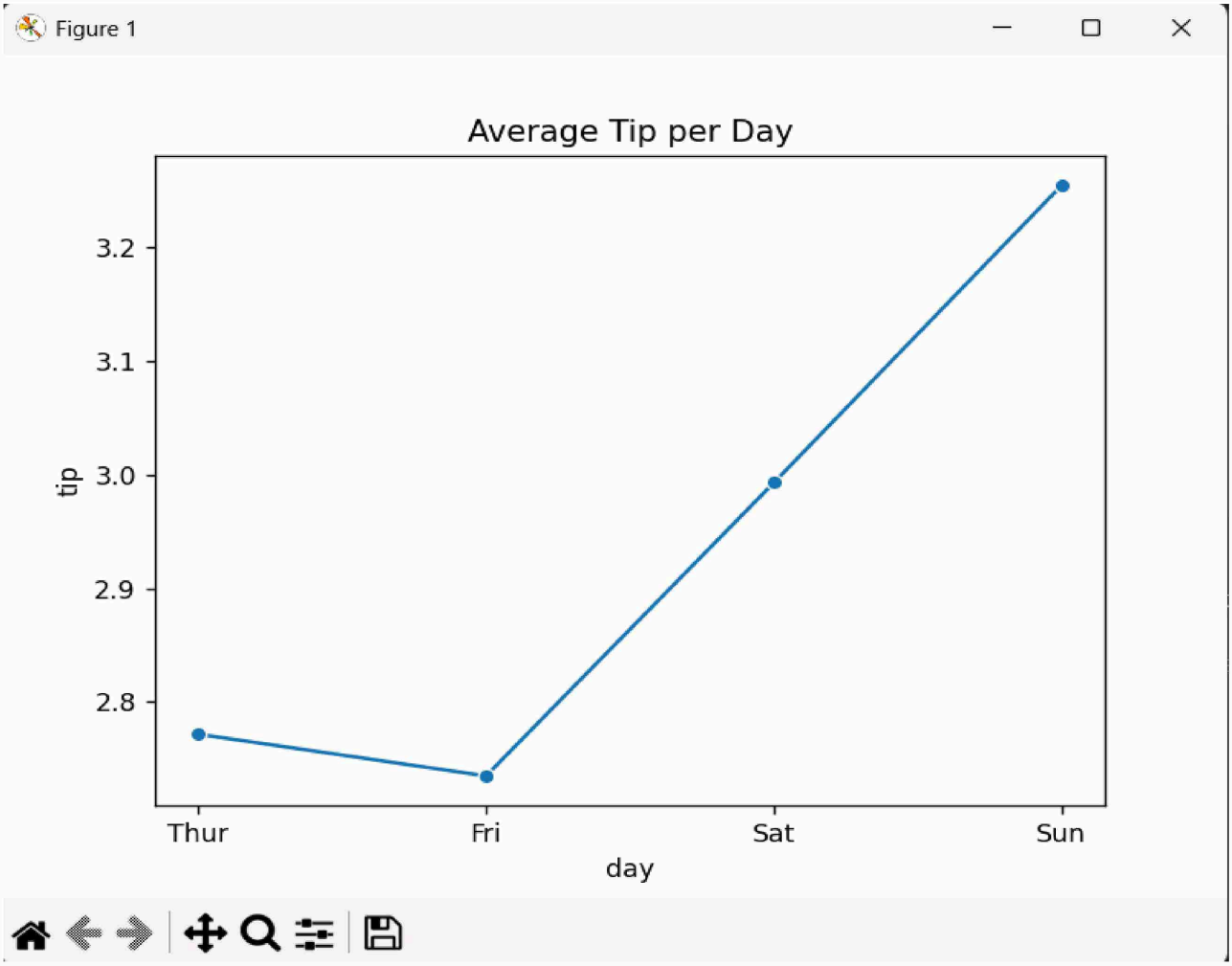


Figure 1
Average Tip per Day
tip
3.2
3.1
3.0
2.9
2.8
Thur
Fri
Sat
Sun
day

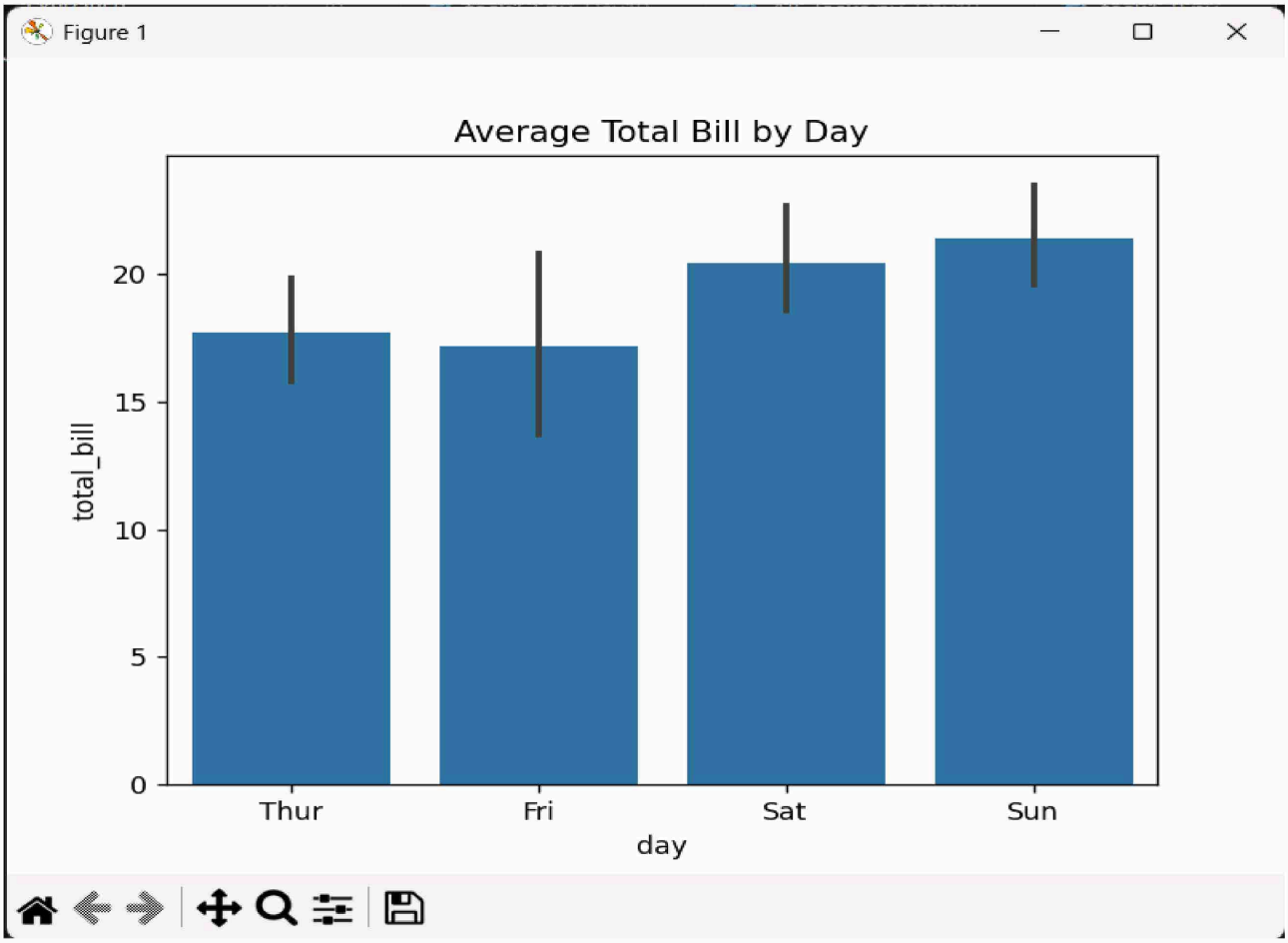


Figure 1
Average Total Bill by Day
total_bill
20
15
10
5
0
Thur
Fri
Sat
Sun
day

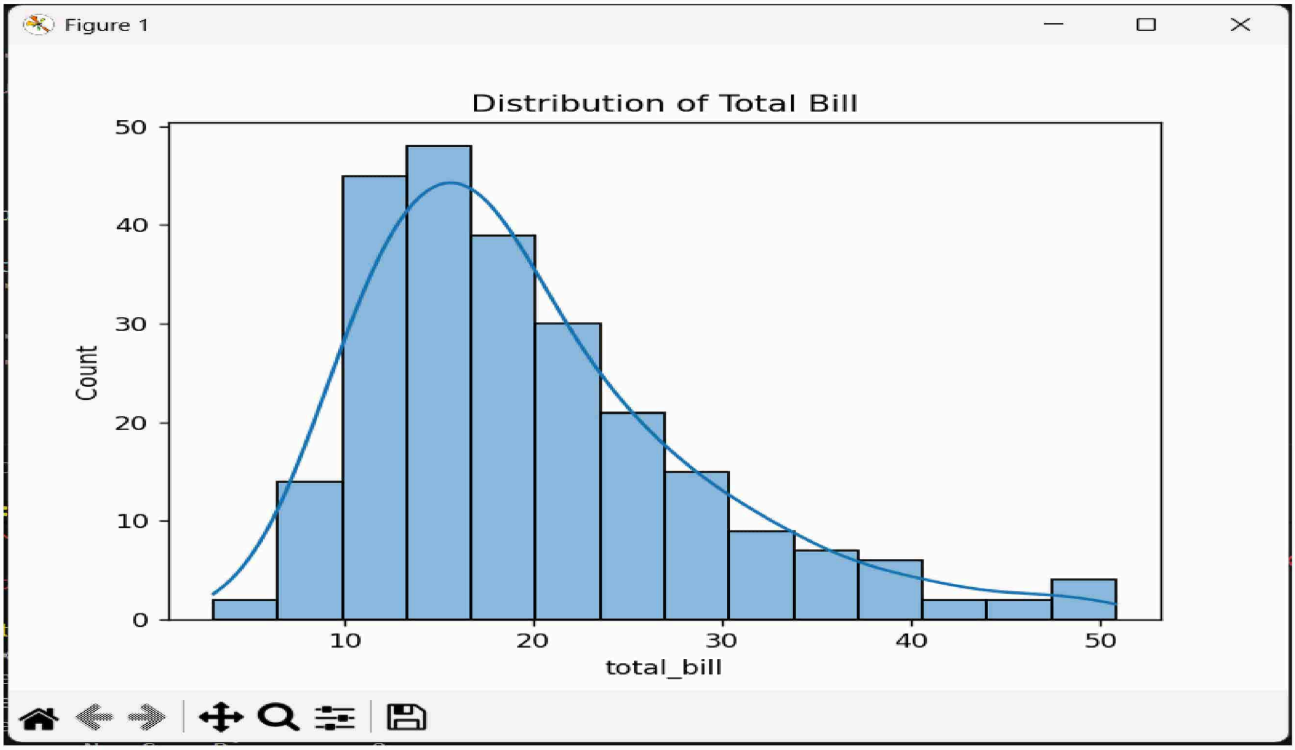


Figure 1
Distribution of Total Bill
Count
total_bill
0
10
20
30
40
50

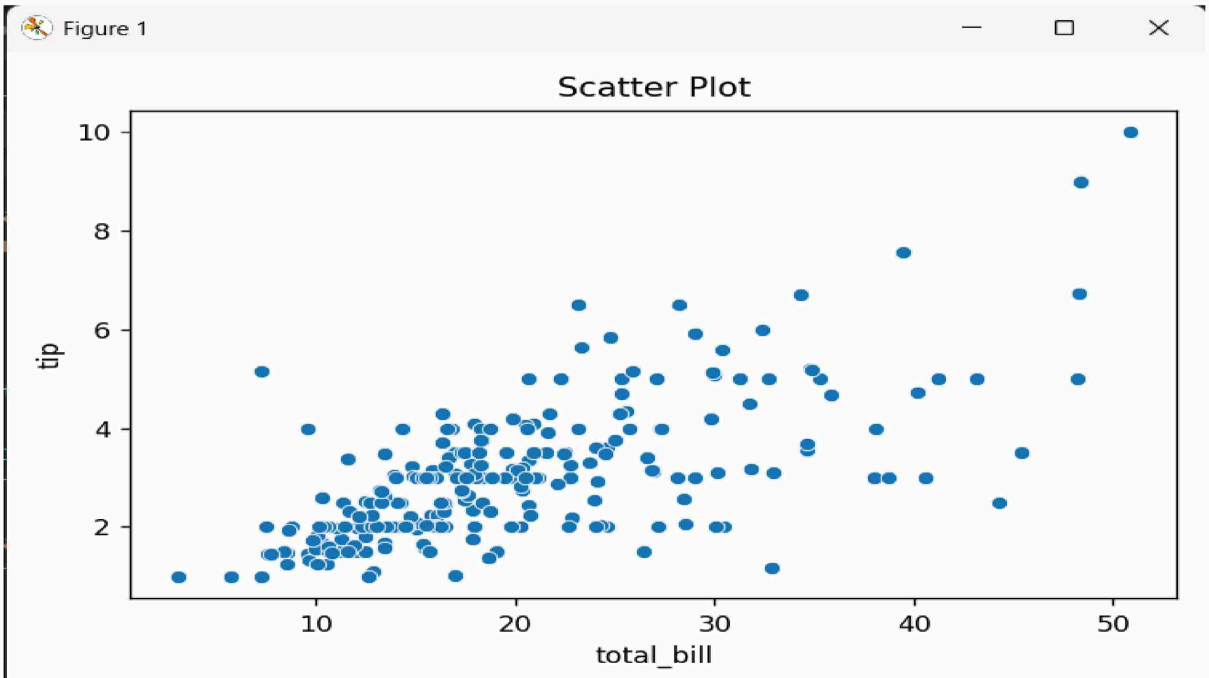
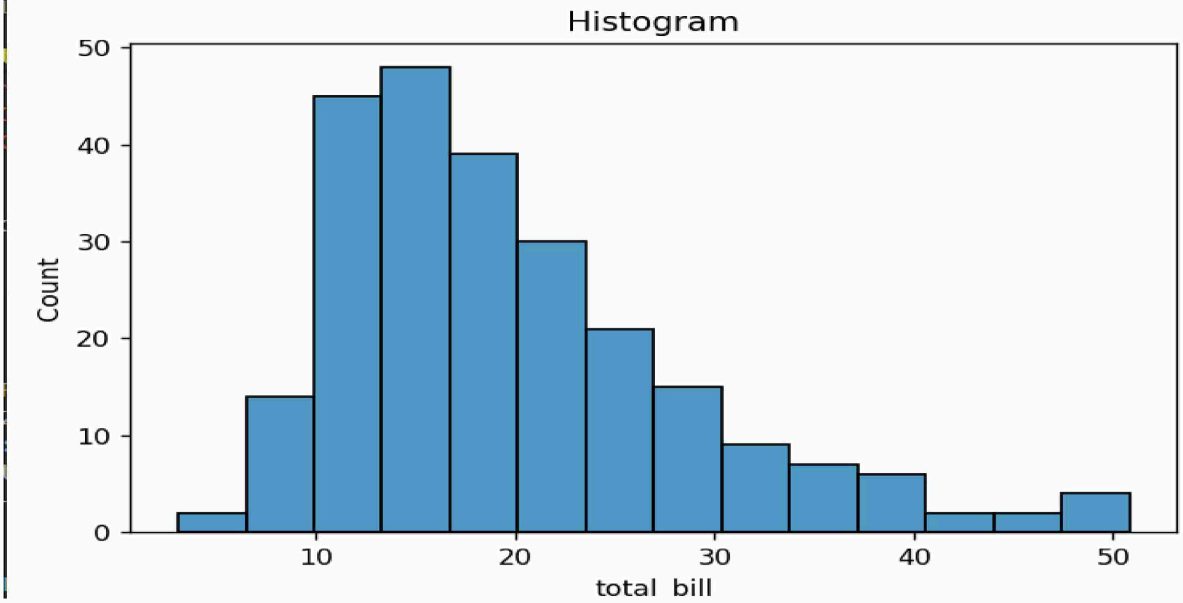


Figure 1
Scatter Plot
tip
total_bill
2
4
6
8
10
10
20
30
40
50


Histogram
Count
total_bill
0
10
20
30
40
50
10
20
30
40
50

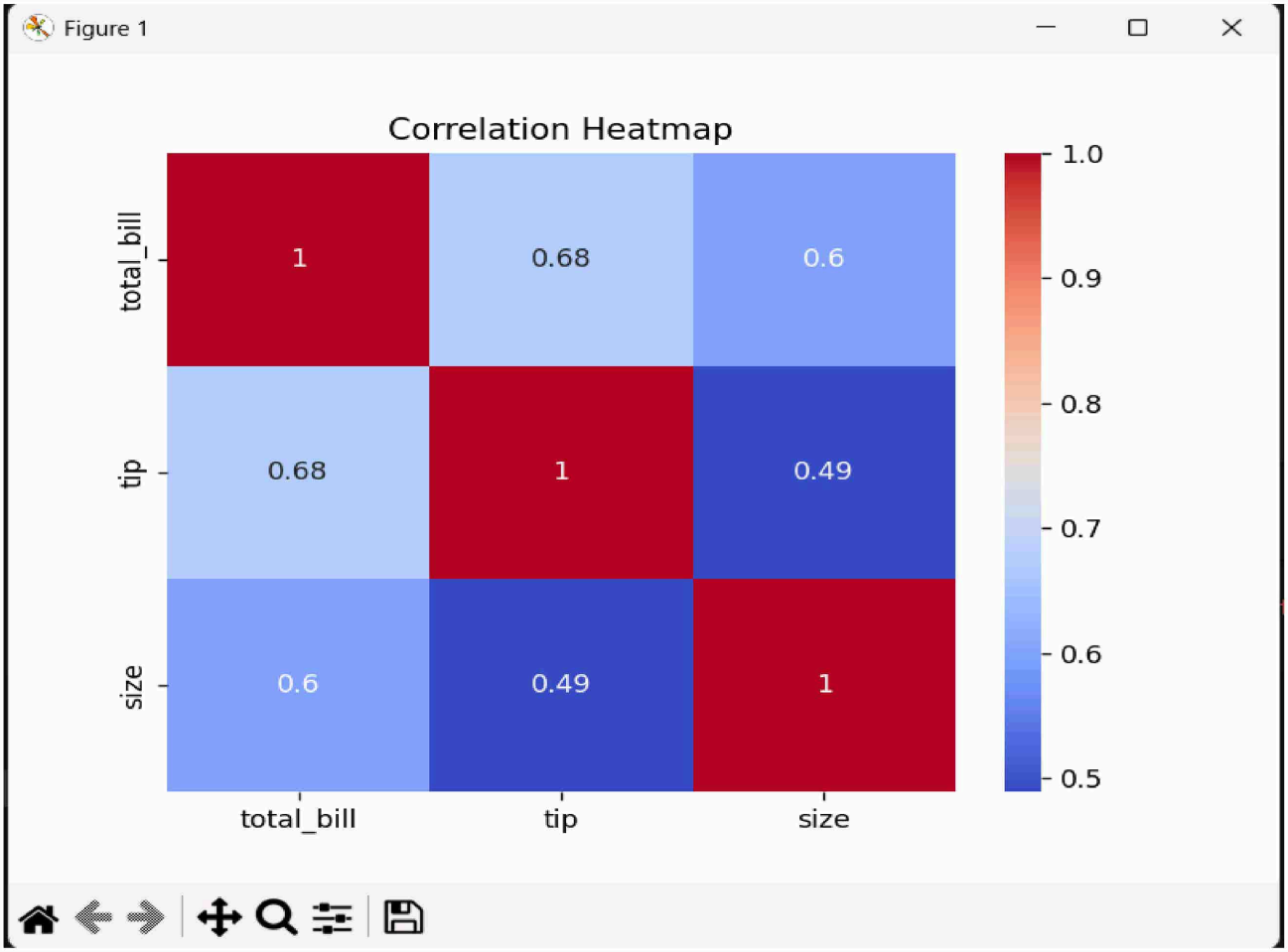


Figure 1
Correlation Heatmap
total_bill
tip
size
1
0.68
0.6
0.68
1
0.49
0.6
0.49
1
total_bill
tip
size
1.0
0.9
0.8
0.7
0.6
0.5

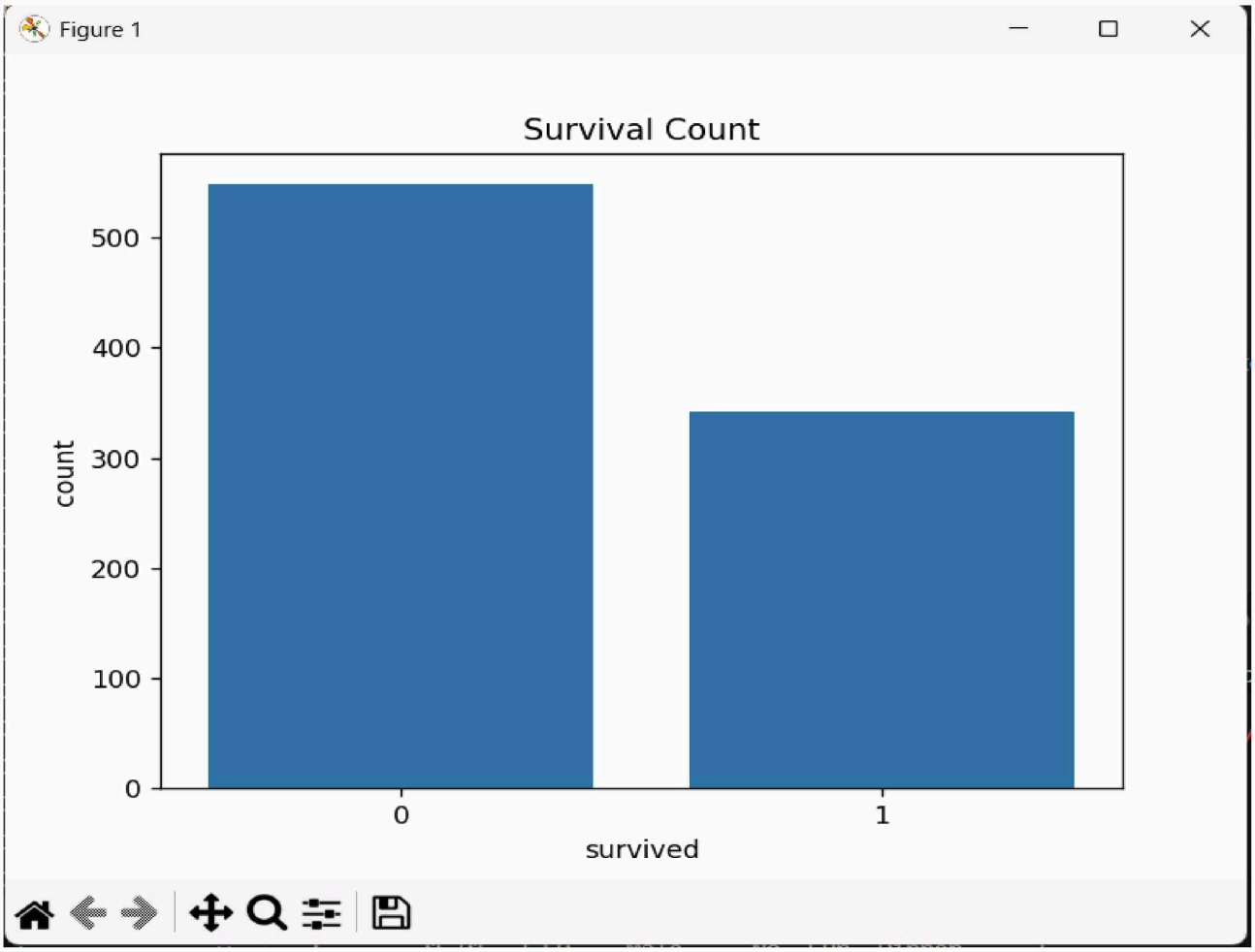


Figure 1
Survival Count
count
500
400
300
200
100
0
0
1
survived

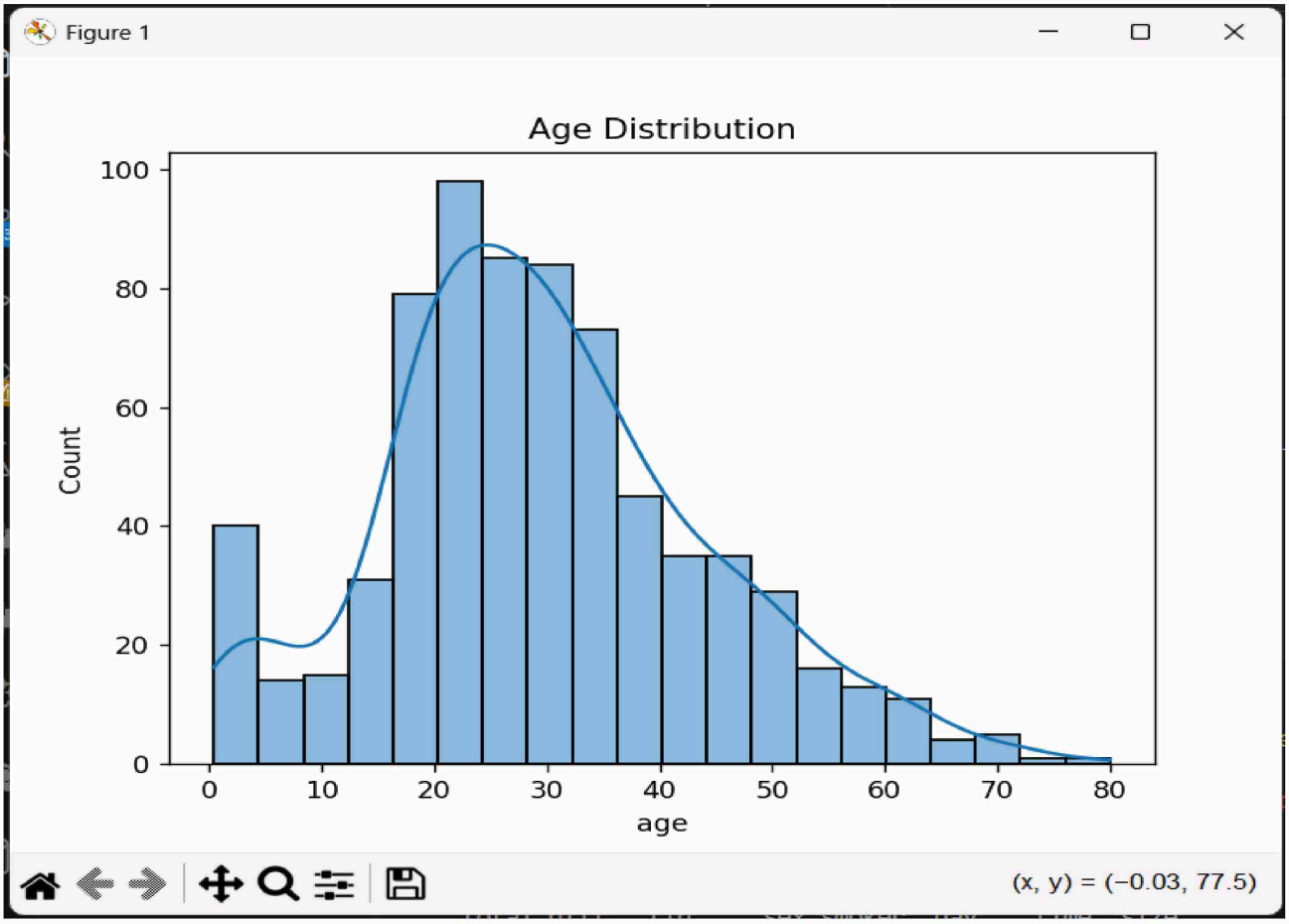


Figure 1
Age Distribution
Count
age
0
20
40
60
80
100
0
10
20
30
40
50
60
70
80
(x, y) = (−0.03, 77.5)

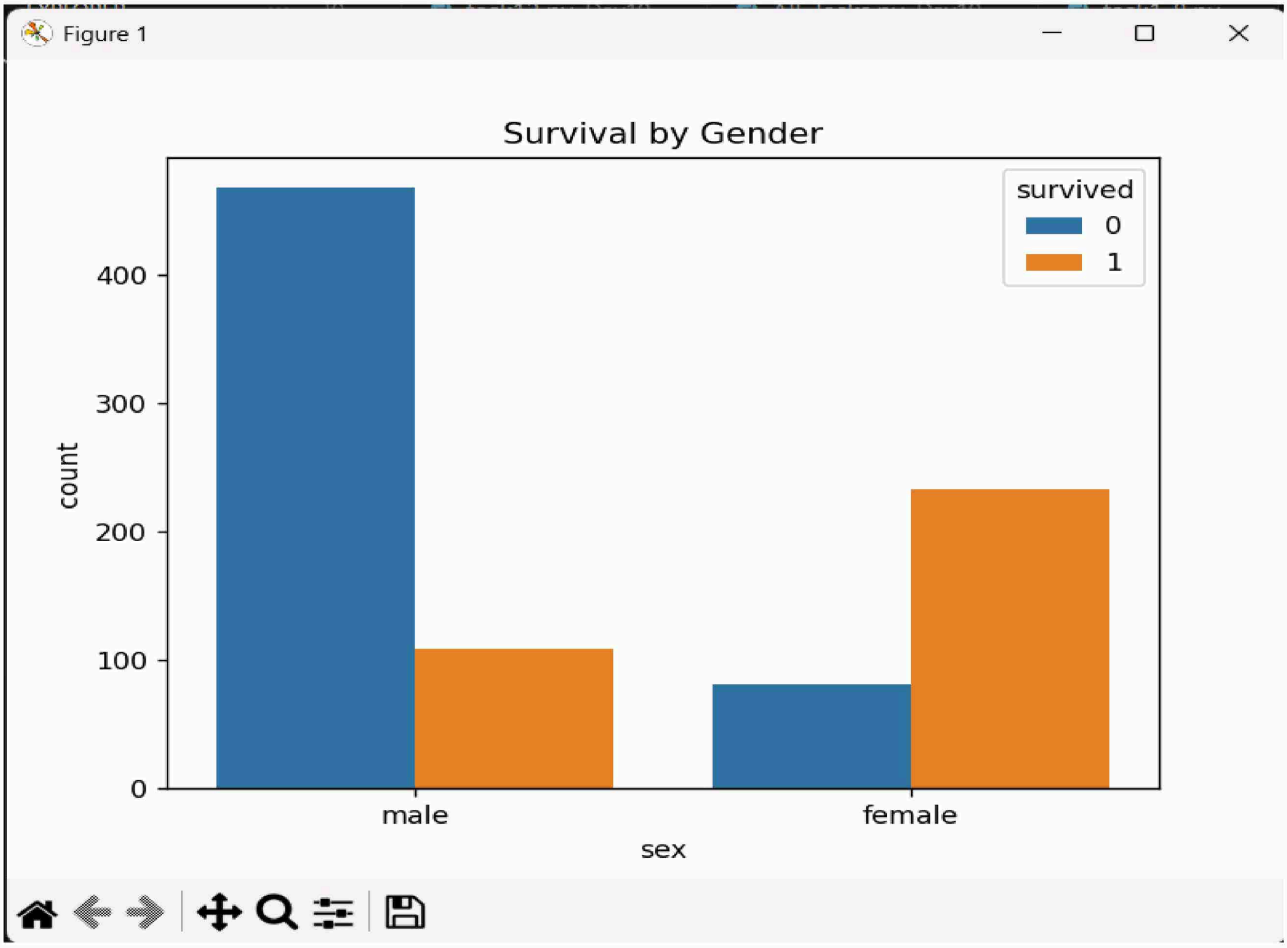


Figure 1
Survival by Gender
survived
0
1
count
sex
0
100
200
300
400
male
female

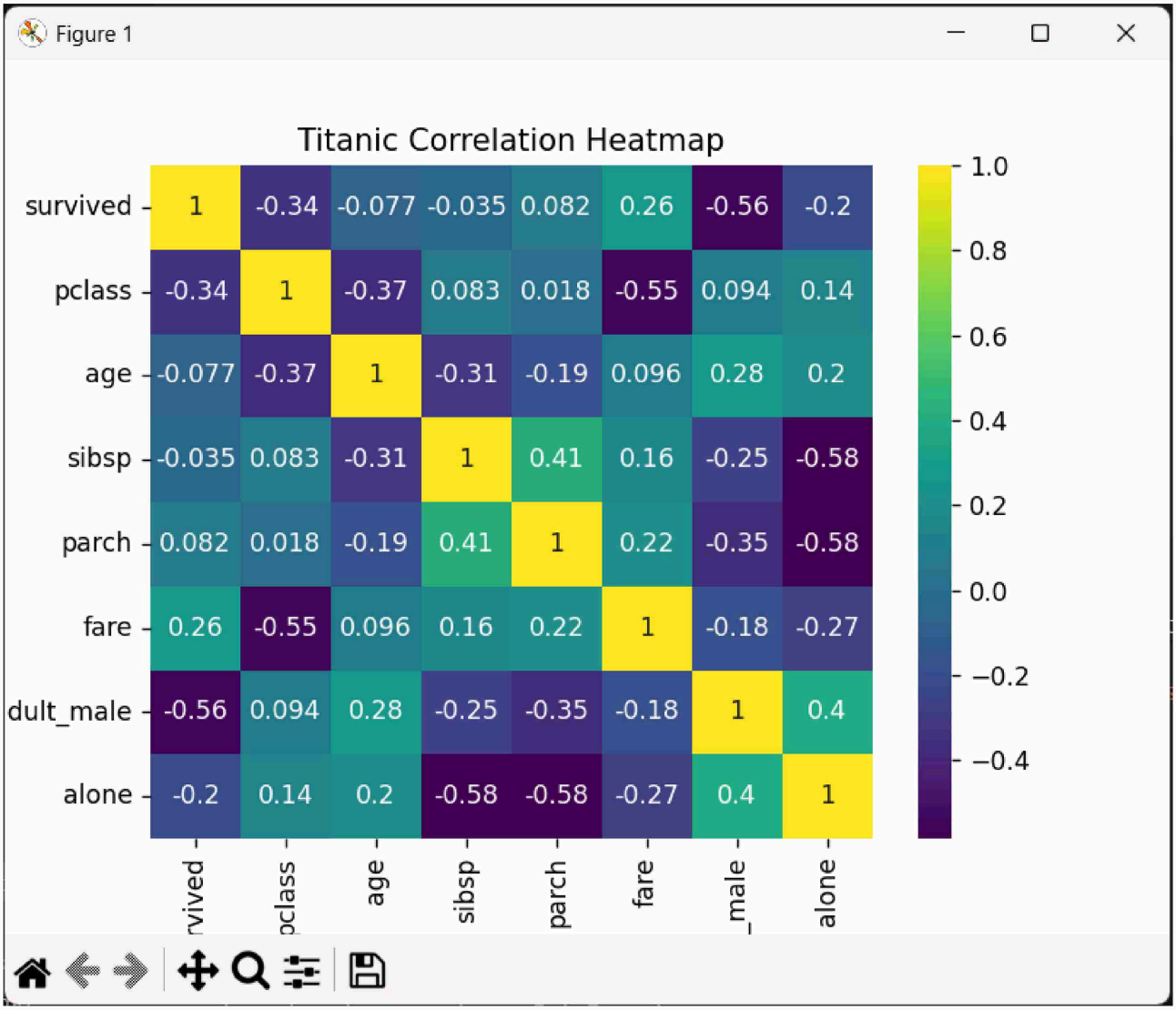
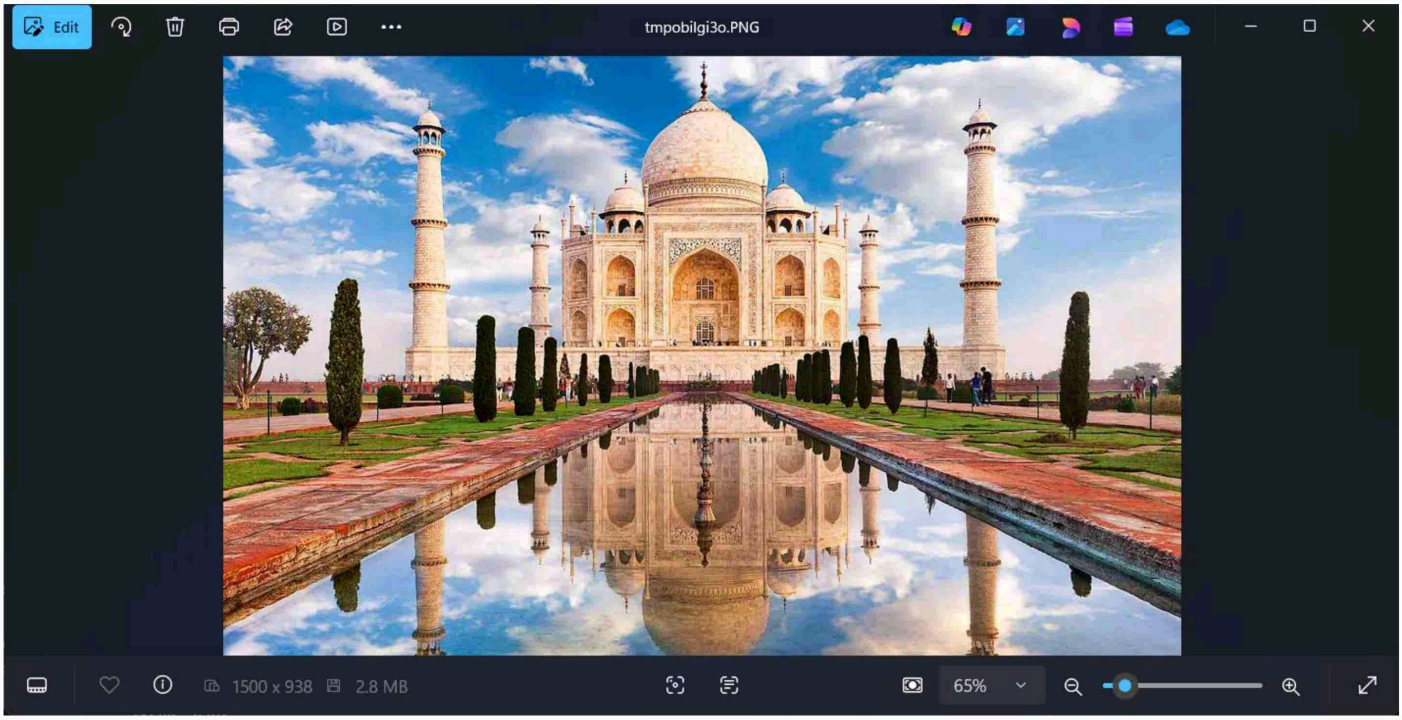


Figure 1
Titanic Correlation Heatmap
survived
pclass
age
sibsp
parch
fare
dult_male
alone
1 -0.34 -0.077 -0.035 0.082 0.26 -0.56 -0.2
-0.34 1 -0.37 0.083 0.018 -0.55 0.094 0.14
-0.077 -0.37 1 -0.31 -0.19 0.096 0.28 0.2
-0.035 0.083 -0.31 1 0.41 0.16 -0.25 -0.58
0.082 0.018 -0.19 0.41 1 0.22 -0.35 -0.58
0.26 -0.55 0.096 0.16 0.22 1 -0.18 -0.27
-0.56 0.094 0.28 -0.25 -0.35 -0.18 1 0.4
-0.2 0.14 0.2 -0.58 -0.58 -0.27 0.4 1
rvived
pclass
age
sibsp
parch
fare
_male
alone
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4

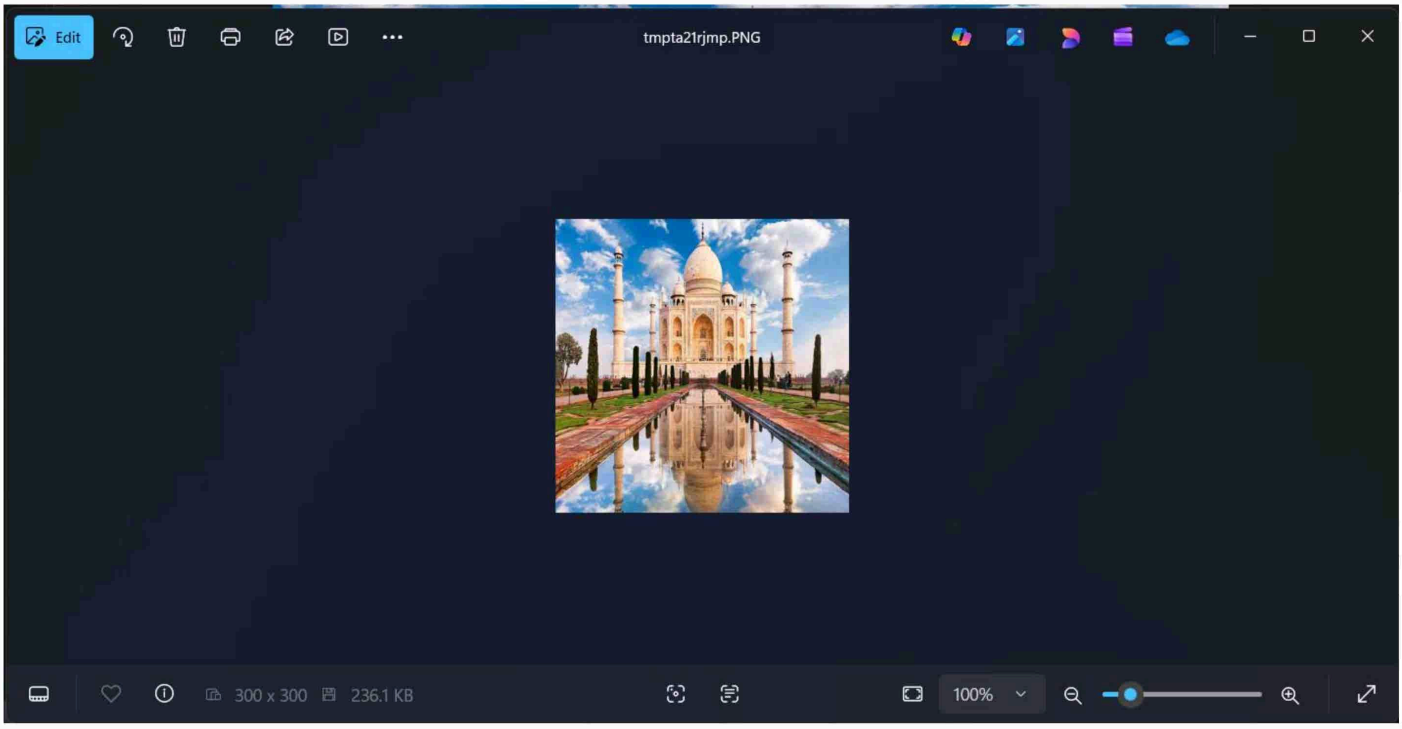
```
from PIL import Image, ImageFilter

# Task1
img = Image.open("image.jpg")
img.show()

# Task2
print(img.size)
print(img.mode)
print(img.format)

# Task3
resized = img.resize((300, 300))
resized.show()

# Task4
gray = img.convert("L")
gray.show()

# Task5
bw = img.convert("1")
bw.show()

# Task6
rotated = img.rotate(45)
rotated.show()

# Task7
cropped = img.crop((50, 50, 300, 300))
cropped.show()

# Task8
gray.save("gray_image.jpg")
```

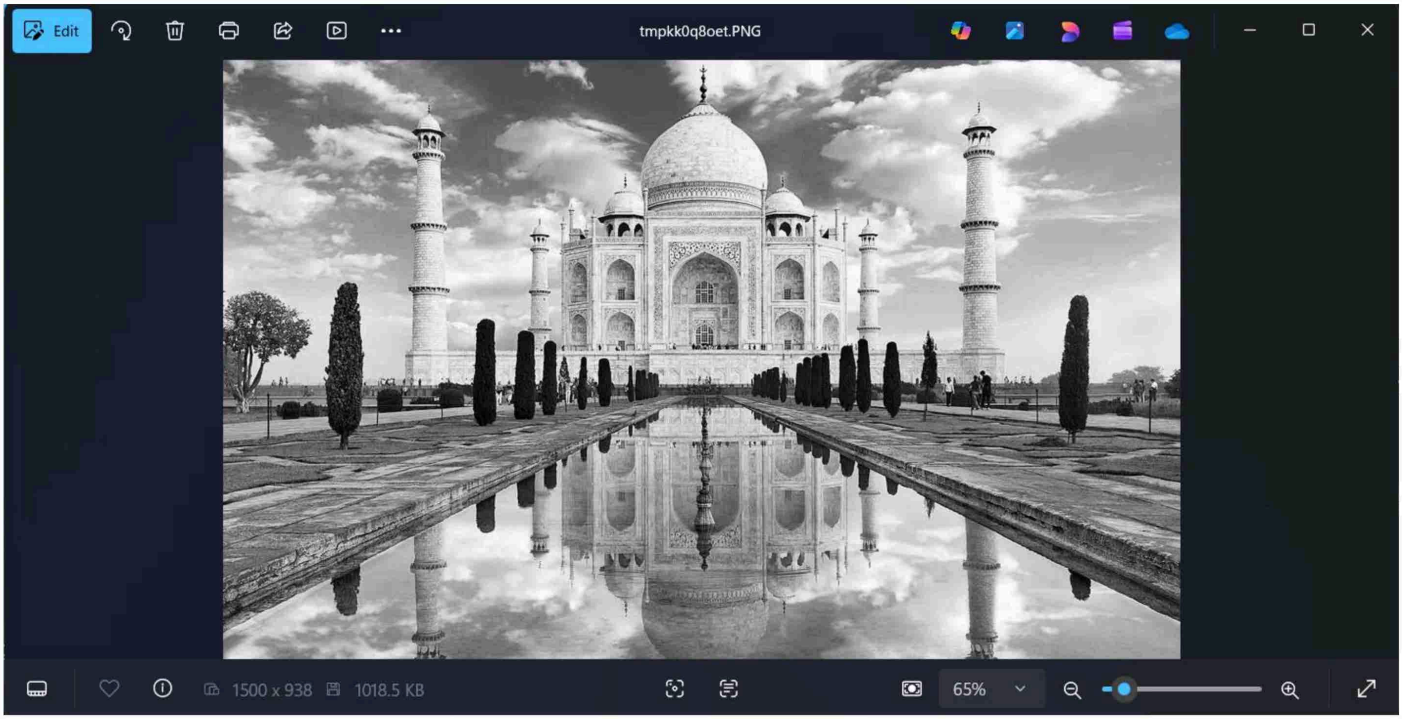
```
# Task9
img.thumbnail((100, 100))
img.show()

# Task10
blur = img.filter(ImageFilter.BLUR)
blur.show()

# Task11
sharp = img.filter(ImageFilter.SHARPEN)
sharp.show()

# Task12
edge = img.filter(ImageFilter.FIND_EDGES)
edge.show()

# Task13
img = Image.open("image.jpg")

img = img.resize((200, 200))
img = img.convert("L")
img = img.rotate(90)

img.save("new_image.jpg")
img.show()
```

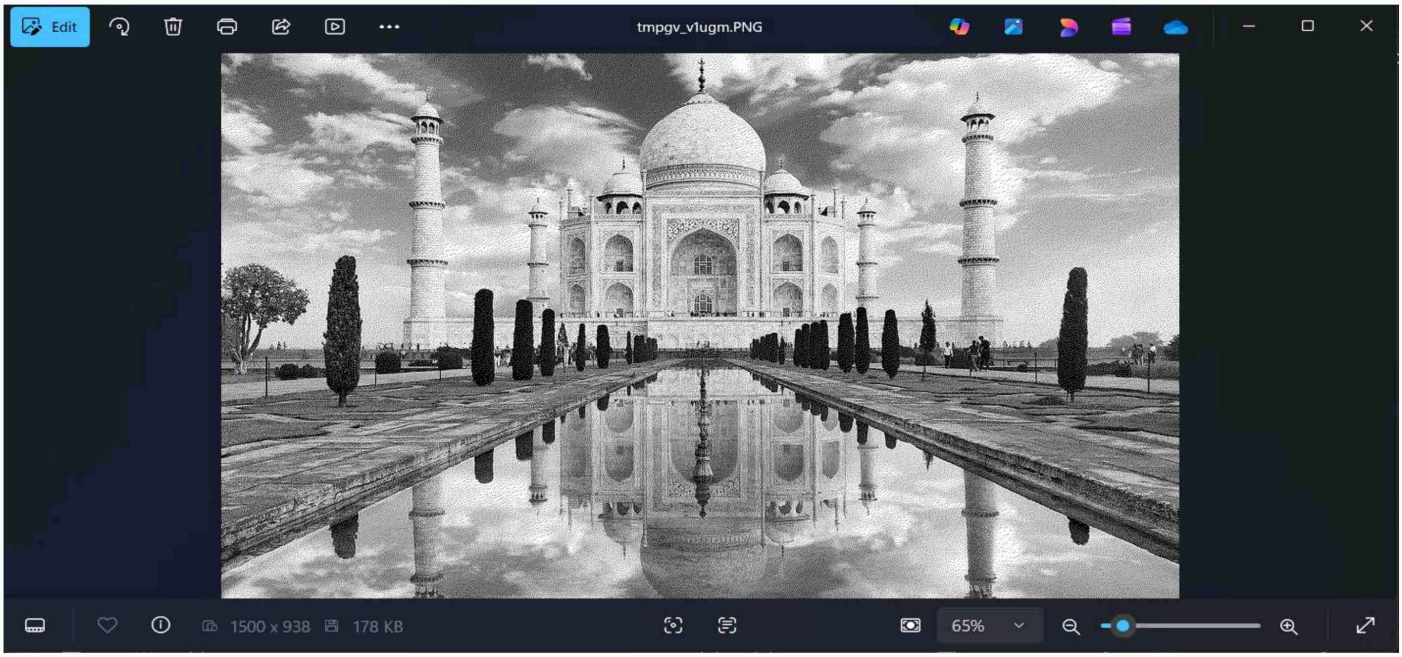
Outputs:




```
PS D:\Internship\Day22> python task1-13.py
(1500, 938)
RGB
JPEG
```




Edit
tmpkk0q8oet.PNG
1500 x 938
1018.5 KB
65%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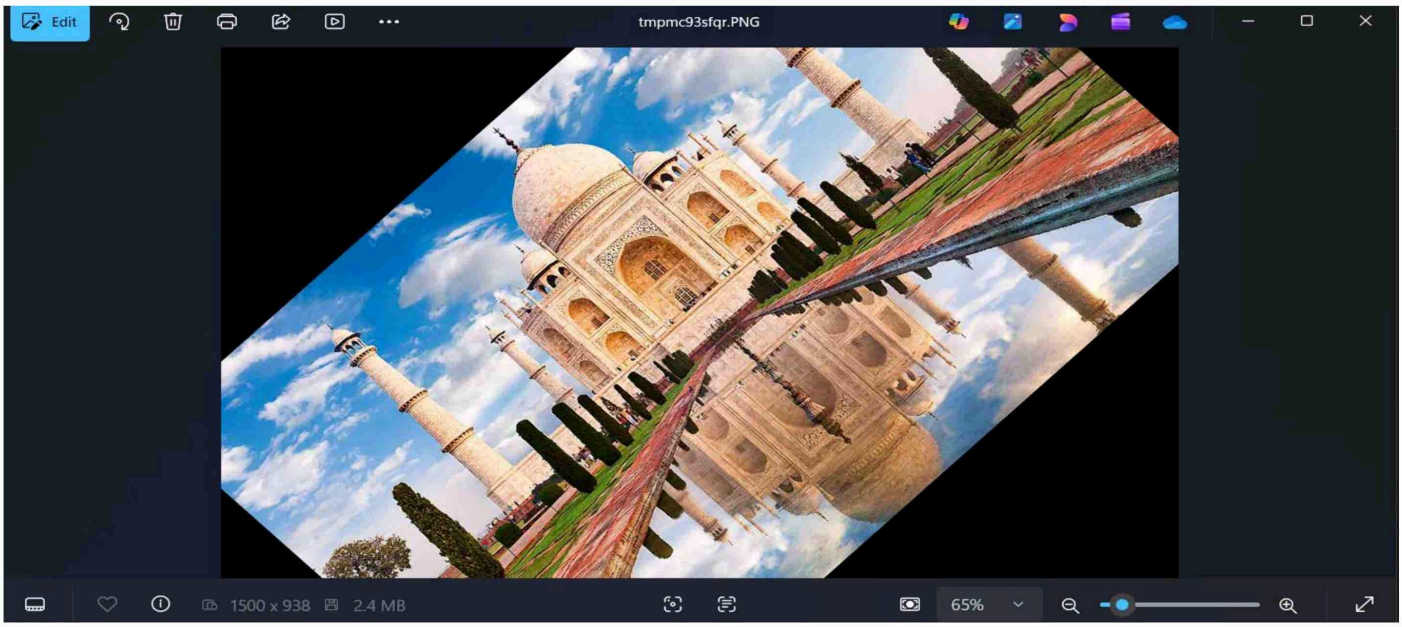
tmpgv_v1ugm.PNG
1500 x 938
178 KB
65%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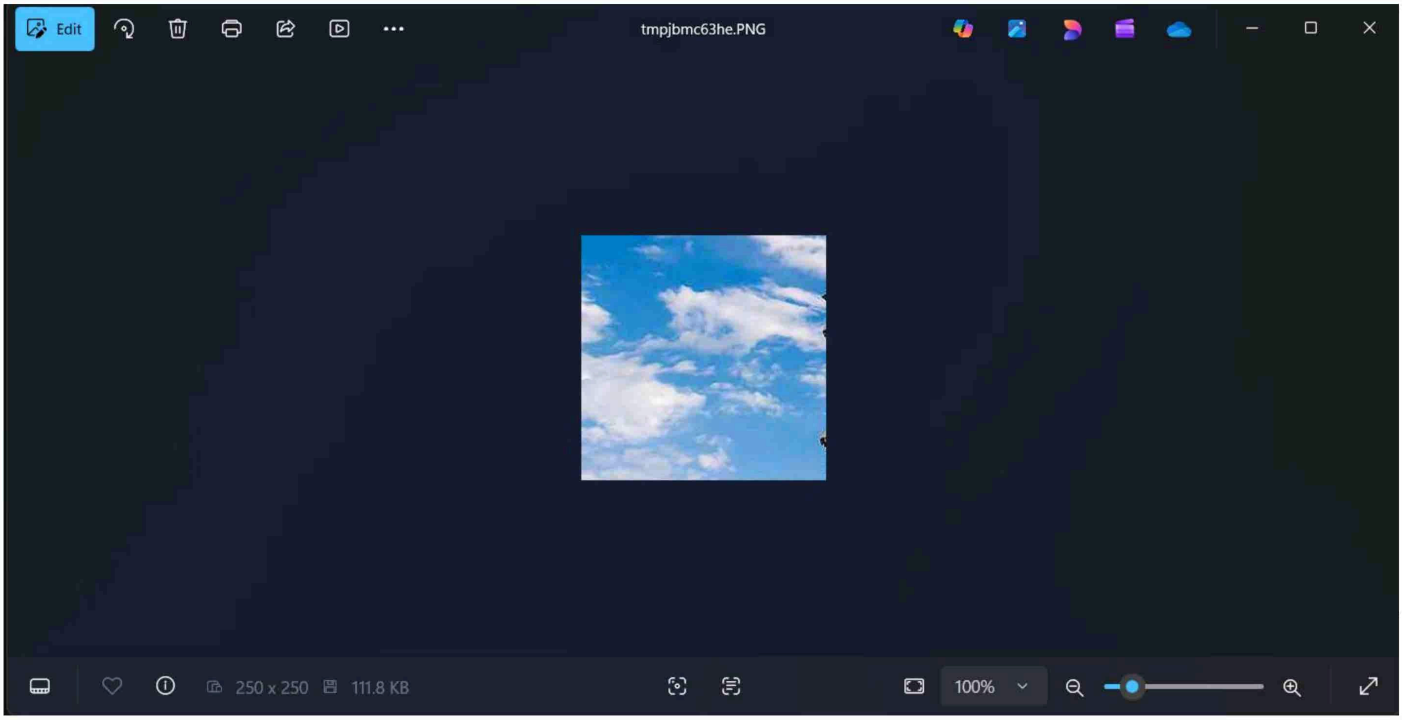
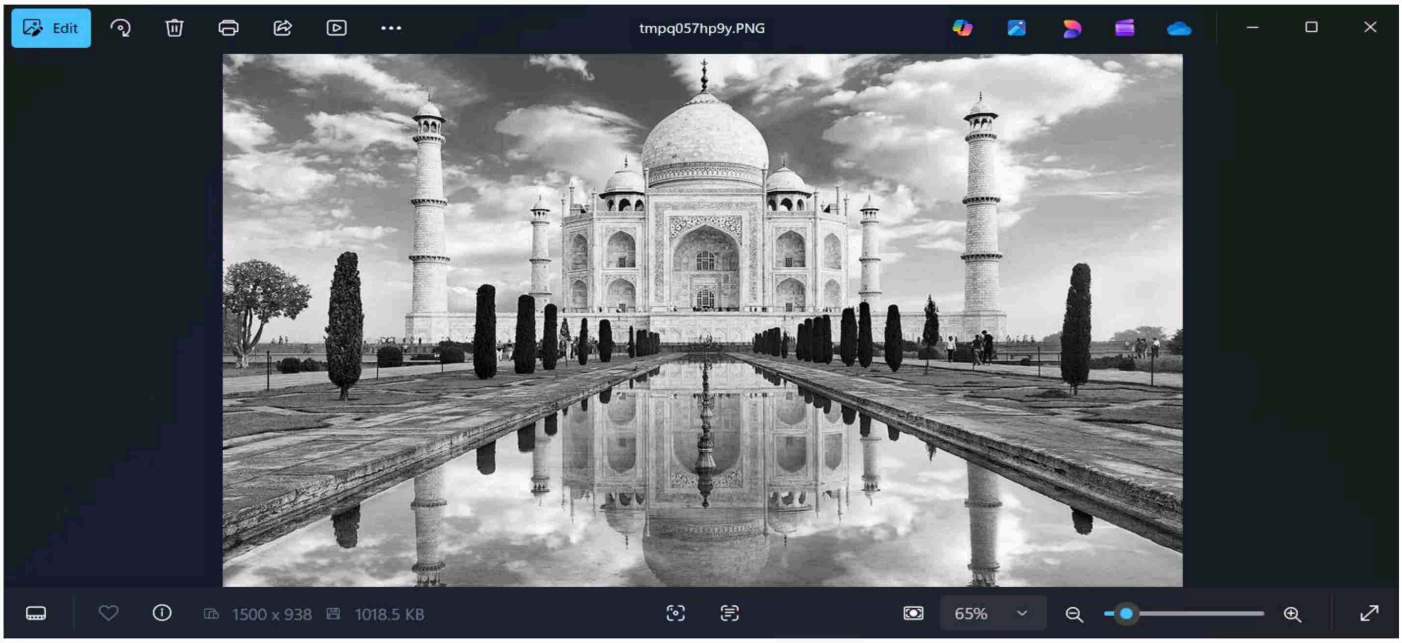
tmpmc93sfqr.PNG
1500 x 938
2.4 MB
65%


Edit
tmpjbmc63he.PNG
250 x 250
111.8 KB
100%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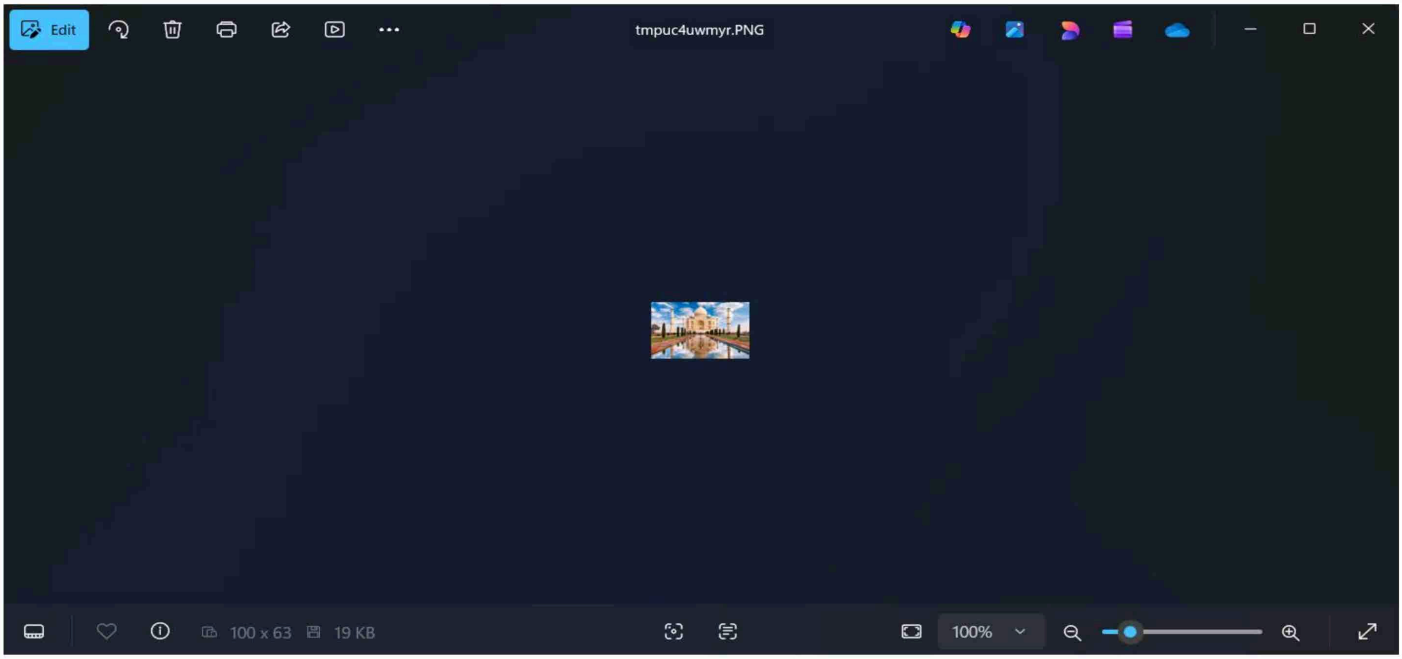
tmpq057hp9y.PNG
1500 x 938
1018.5 KB
65%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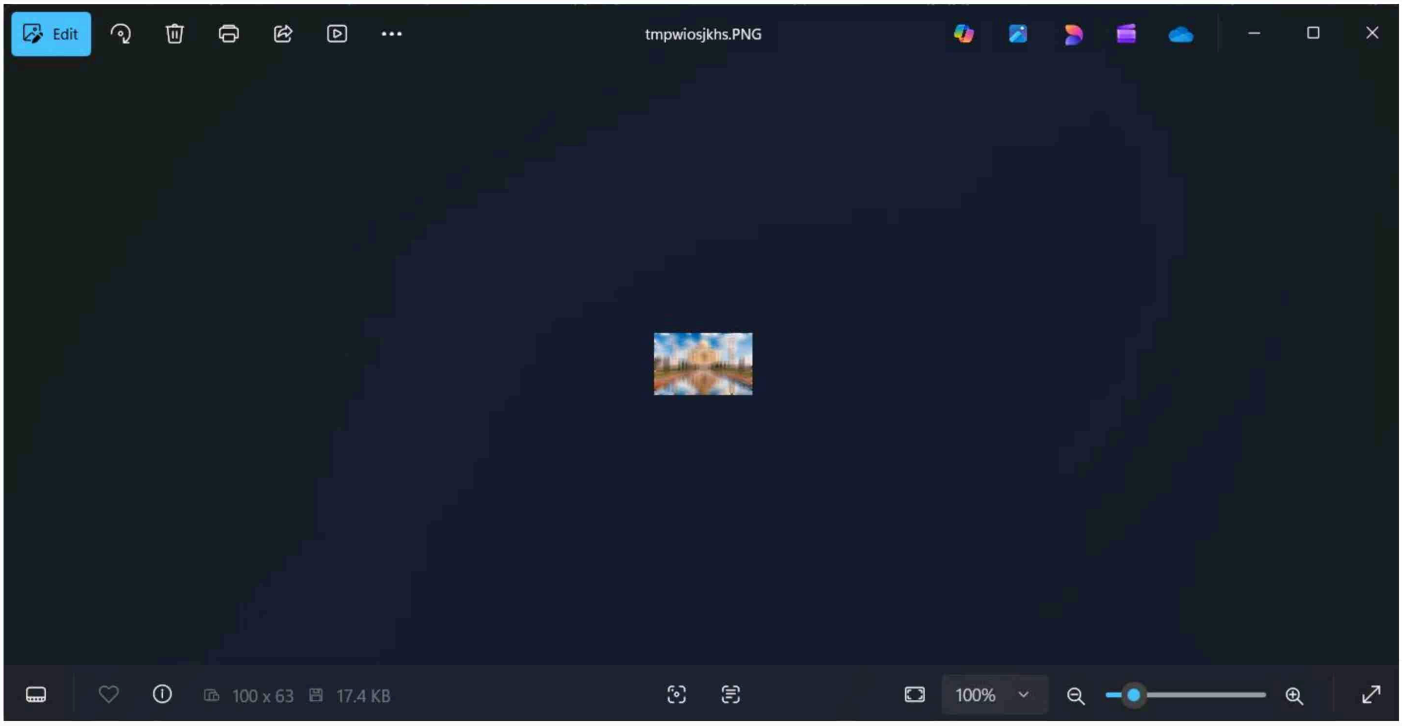
tmpuc4uwmyr.PNG
100 x 63
19 KB
100%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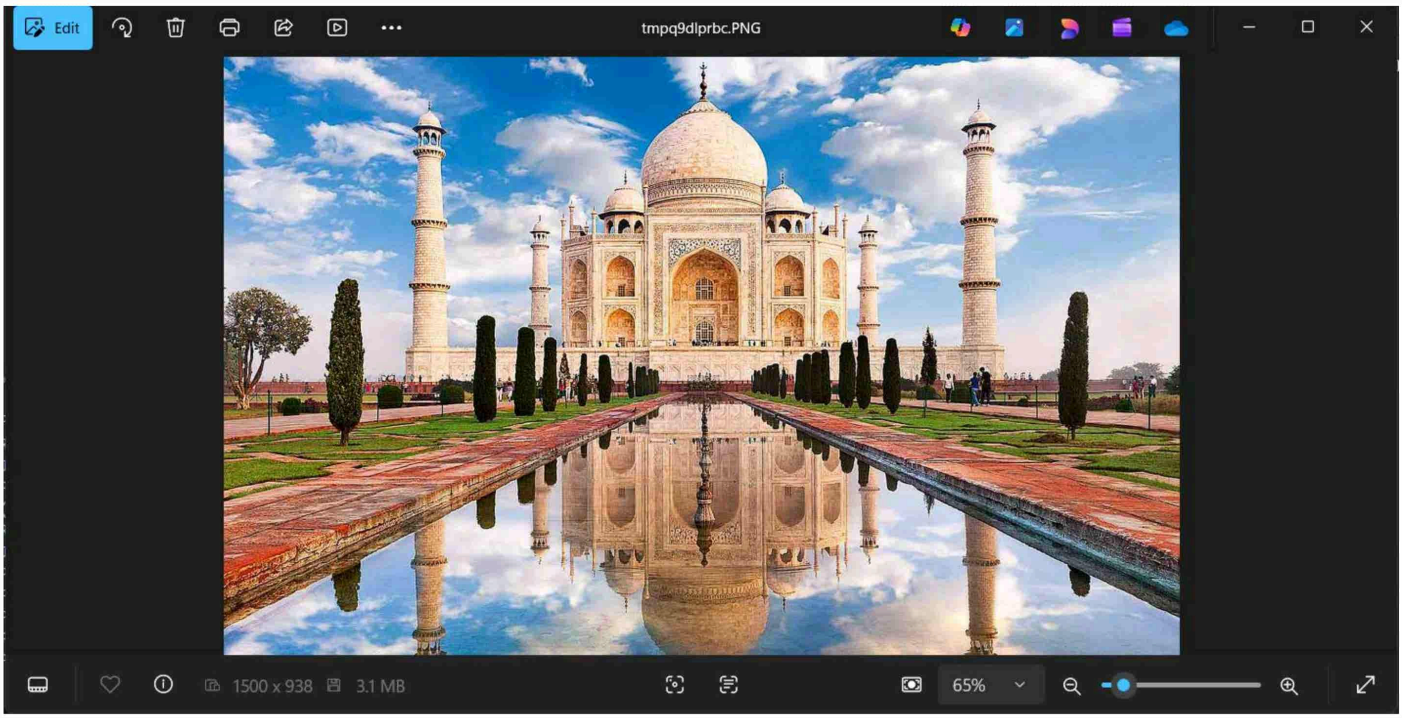
tmpwiosjkhs.PNG
100 x 63
17.4 KB
100%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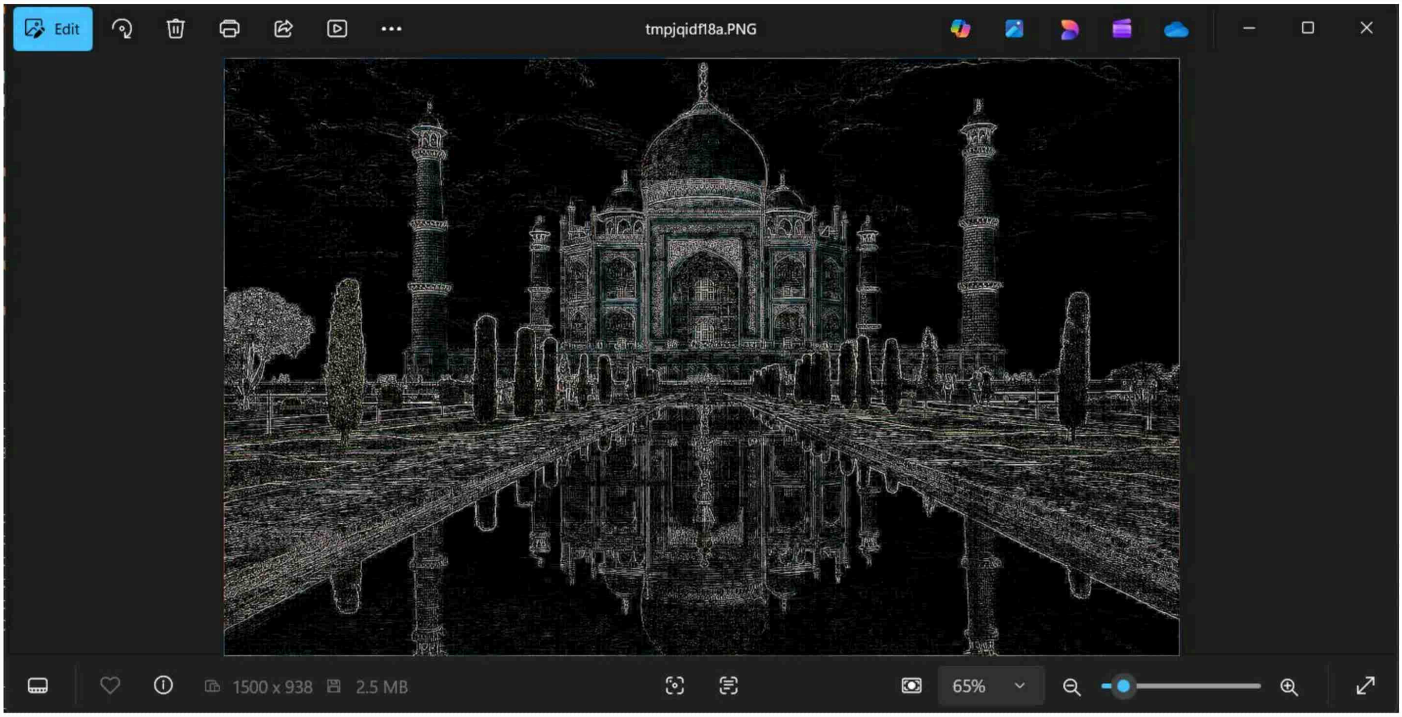
tmpq9dlprbc.PNG
1500 x 938
3.1 MB
65%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2.5 MB
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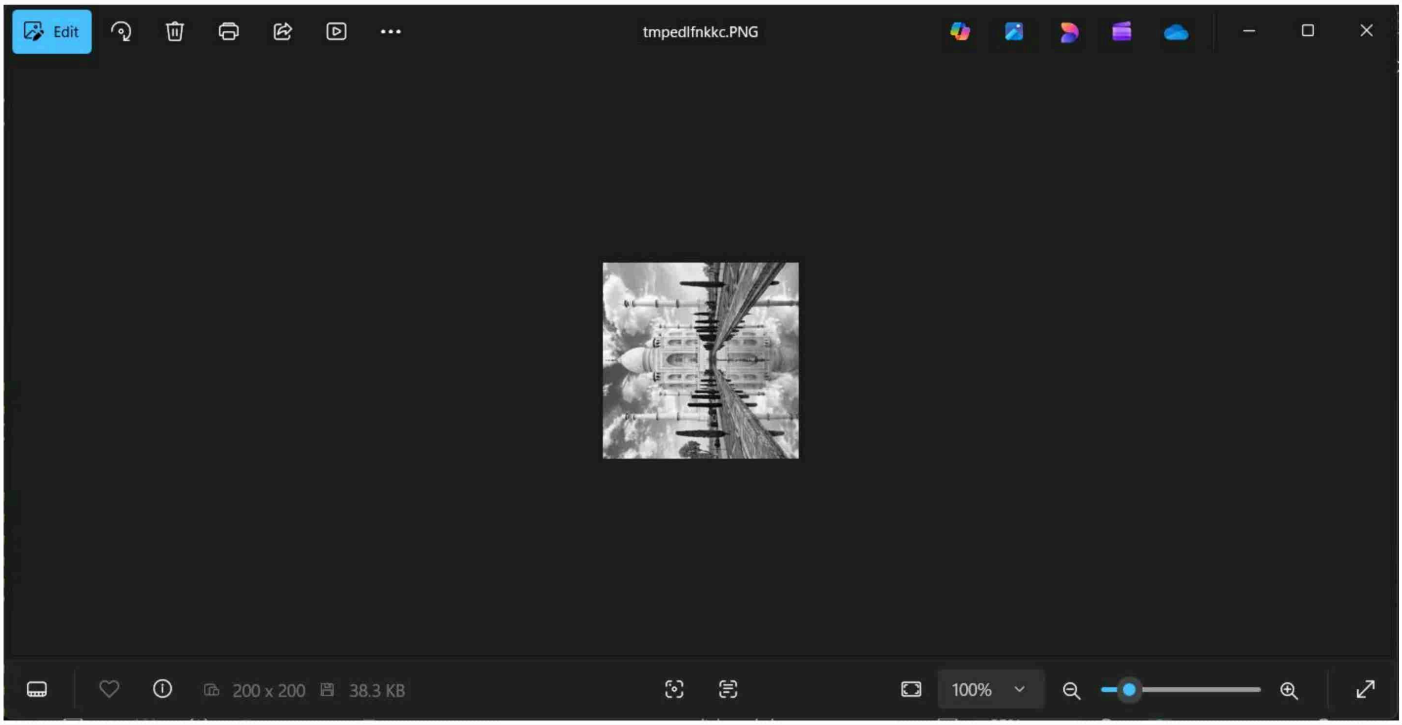
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200 x 200
38.3 KB
100%